

## SAFETY DATA SHEET

Creation Date 09-May-2012

Revision Date 26-March-2024

Revision Number 3

### 1. Identification

**Product Name** Zinc wire

**Cat No. :** S42705

**CAS-No** 7440-66-6  
**Synonyms** No information available

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

##### **Importer/Distributor**

Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11

Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99

**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

**WHMIS 2015 Classification** Not classified under the Hazardous Products Regulations (SOR/2015-17)

Based on available data, the classification criteria are not met

#### Label Elements

None required

### 3. Composition/Information on Ingredients

| Component | CAS-No    | Weight % |
|-----------|-----------|----------|
| Zinc      | 7440-66-6 | <= 100   |

#### 4. First-aid measures

|                                                               |                                                                                                                         |
|---------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>                                            | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.         |
| <b>Skin Contact</b>                                           | Wash off immediately with plenty of water for at least 15 minutes. Get medical attention immediately if symptoms occur. |
| <b>Inhalation</b>                                             | Remove to fresh air. If breathing is difficult, give oxygen. Get medical attention immediately if symptoms occur.       |
| <b>Ingestion</b>                                              | Do NOT induce vomiting. Get medical attention.                                                                          |
| <b>Most important symptoms/effects<br/>Notes to Physician</b> | None reasonably foreseeable.<br>Treat symptomatically                                                                   |

#### 5. Fire-fighting measures

|                                         |                          |
|-----------------------------------------|--------------------------|
| <b>Unsuitable Extinguishing Media</b>   | Water                    |
| <b>Flash Point</b>                      | No information available |
| <b>Method -</b>                         | No information available |
| <b>Autoignition Temperature</b>         | 460 °C / 860 °F          |
| <b>Explosion Limits</b>                 |                          |
| <b>Upper</b>                            | No data available        |
| <b>Lower</b>                            | No data available        |
| <b>Sensitivity to Mechanical Impact</b> | No information available |
| <b>Sensitivity to Static Discharge</b>  | No information available |

##### Specific Hazards Arising from the Chemical

Thermal decomposition can lead to release of irritating gases and vapors. Keep product and empty container away from heat and sources of ignition.

##### Hazardous Combustion Products

Nitrogen oxides (NO<sub>x</sub>). Sulfur oxides. Ammonia.

##### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

##### NFPA

**Health**  
1

**Flammability**  
1

**Instability**  
0

**Physical hazards**  
N/A

#### 6. Accidental release measures

|                                             |                                                                                                                                                              |
|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Use personal protective equipment as required. Ensure adequate ventilation. Avoid dust formation.                                                            |
| <b>Environmental Precautions</b>            | Should not be released into the environment. Do not flush into surface water or sanitary sewer system. See Section 12 for additional Ecological Information. |
| <b>Methods for Containment and Clean Up</b> | Sweep up and shovel into suitable containers for disposal. Avoid dust formation.                                                                             |

#### 7. Handling and storage

|                 |                                                                                                                                                                                   |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid contact with skin, eyes or clothing. Avoid ingestion and inhalation. Avoid dust formation. |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**Storage.** Keep containers tightly closed in a dry, cool and well-ventilated place. Incompatible Materials. Strong oxidizing agents. Strong acids. Strong bases. Amines. Combustible material. Peroxides. Metals.

## 8. Exposure controls / personal protection

**Exposure Guidelines** This product does not contain any hazardous materials with occupational exposure limitsestablished by the region specific regulatory bodies.

**Engineering Measures** None under normal use conditions.

### Personal protective equipment

**Eye Protection** Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.

**Hand Protection** Protective gloves

| Glove material | Breakthrough time | Glove thickness | Glove comments         |
|----------------|-------------------|-----------------|------------------------|
| Natural rubber | See manufacturers | -               | Splash protection only |
| Nitrile rubber | recommendations   |                 |                        |
| Neoprene       |                   |                 |                        |
| PVC            |                   |                 |                        |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

**Respiratory Protection**  
No protective equipment is needed under normal use conditions.

**Recommended Filter type:** Particle filter

### Environmental exposure controls

Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

## 9. Physical and chemical properties

|                          |                          |
|--------------------------|--------------------------|
| Physical State           | Solid; Various Form      |
| Appearance               | Grey                     |
| Odor                     | No information available |
| Odor Threshold           | No information available |
| pH                       | No information available |
| Melting Point/Range      | 419 °C / 786.2 °F        |
| Boiling Point/Range      | 907 °C / 1664.6 °F       |
| Flash Point              | No information available |
| Evaporation Rate         | Not applicable           |
| Flammability (solid,gas) | No information available |

**Flammability or explosive limits**

|                                        |                          |
|----------------------------------------|--------------------------|
| Upper                                  | No data available        |
| Lower                                  | No data available        |
| Vapor Pressure                         | 1.3 mbar @ 478 °C        |
| Vapor Density                          | Not applicable           |
| Specific Gravity                       | 7.140                    |
| Solubility                             | Insoluble in water       |
| Partition coefficient; n-octanol/water | No data available        |
| Autoignition Temperature               | 460 °C / 860 °F          |
| Decomposition Temperature              | No information available |
| Viscosity                              | Not applicable           |
| Molecular Formula                      | Zn                       |
| Molecular Weight                       | 65.36                    |

## 10. Stability and reactivity

|                                         |                                                                                                      |
|-----------------------------------------|------------------------------------------------------------------------------------------------------|
| <b>Reactive Hazard</b>                  | None known, based on information available                                                           |
| <b>Stability</b>                        | Stable under normal conditions.                                                                      |
| <b>Conditions to Avoid</b>              | Incompatible products. Excess heat. Avoid dust formation.                                            |
| <b>Incompatible Materials</b>           | Strong oxidizing agents, Strong acids, Strong bases, Amines, Combustible material, Peroxides, Metals |
| <b>Hazardous Decomposition Products</b> | Nitrogen oxides (NOx), Sulfur oxides, Ammonia                                                        |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                                                             |
| <b>Hazardous Reactions</b>              | None under normal processing.                                                                        |

## 11. Toxicological information

**Acute Toxicity**

**Product Information** No acute toxicity information is available for this product

**Component Information**

| Component | LD50 Oral                | LD50 Dermal | LC50 Inhalation |
|-----------|--------------------------|-------------|-----------------|
| Zinc      | LD50 = 630 mg/kg ( Rat ) | Not listed  | Not listed      |

**Toxicologically Synergistic Products** No information available

**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

|                        |                                                                                          |
|------------------------|------------------------------------------------------------------------------------------|
| <b>Irritation</b>      | No information available                                                                 |
| <b>Sensitization</b>   | No information available                                                                 |
| <b>Carcinogenicity</b> | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component | CAS-No    | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|-----------|-----------|------------|------------|------------|------------|------------|
| Zinc      | 7440-66-6 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** None known

|                                                   |                                                     |
|---------------------------------------------------|-----------------------------------------------------|
| <b>STOT - repeated exposure</b>                   | None known                                          |
| <b>Aspiration hazard</b>                          | No information available                            |
| <b>Symptoms / effects, both acute and delayed</b> | No information available                            |
| <b>Endocrine Disruptor Information</b>            | No information available                            |
| <b>Other Adverse Effects</b>                      | See actual entry in RTECS for complete information. |

## 12. Ecological information

### Ecotoxicity

Contains a substance which is: Very toxic to aquatic organisms. The product contains following substances which are hazardous for the environment.

| Component | Freshwater Algae                                                                                                                               | Freshwater Fish                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Microtox   | Water Flea                                           |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|------------------------------------------------------|
| Zinc      | EC50: 0.09 - 0.125 mg/L, 72h static (Pseudokirchneriella subcapitata)<br>EC50: 0.11 - 0.271 mg/L, 96h static (Pseudokirchneriella subcapitata) | LC50: = 0.41 mg/L, 96h static (Oncorhynchus mykiss)<br>LC50: = 0.59 mg/L, 96h semi-static (Oncorhynchus mykiss)<br>LC50: 2.16 - 3.05 mg/L, 96h flow-through (Pimephales promelas)<br>LC50: 0.211 - 0.269 mg/L, 96h semi-static (Pimephales promelas)<br>LC50: = 2.66 mg/L, 96h static (Pimephales promelas)<br>LC50: = 30 mg/L, 96h (Cyprinus carpio)<br>LC50: = 0.45 mg/L, 96h semi-static (Cyprinus carpio)<br>LC50: = 7.8 mg/L, 96h static (Cyprinus carpio)<br>LC50: = 0.24 mg/L, 96h flow-through (Oncorhynchus mykiss)<br>LC50: = 3.5 mg/L, 96h static (Lepomis macrochirus) | Not listed | EC50: 0.139 - 0.908 mg/L, 48h Static (Daphnia magna) |

|                                      |                                                                       |
|--------------------------------------|-----------------------------------------------------------------------|
| <b>Persistence and Degradability</b> | Insoluble in water                                                    |
| <b>Bioaccumulation/ Accumulation</b> | No information available.                                             |
| <b>Mobility</b>                      | Is not likely mobile in the environment due its low water solubility. |

## 13. Disposal considerations

|                               |                                                                                                                                                                                                                                                                 |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste Disposal Methods</b> | Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification. |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 14. Transport information

|                 |               |
|-----------------|---------------|
| <b>DOT</b>      | Not regulated |
| <b>TDG</b>      | Not regulated |
| <b>IATA</b>     | Not regulated |
| <b>IMDG/IMO</b> | Not regulated |

## 15. Regulatory information

## International Inventories

| Component | CAS-No    | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|-----------|-----------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| Zinc      | 7440-66-6 | X   | -    | X    | ACTIVE                                        | 231-175-3 | -      | -   |

| Component | CAS-No    | IECSC | KECL     | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|-----------|-----------|-------|----------|------|------|------|------|-------|-------|
| Zinc      | 7440-66-6 | X     | KE-35518 | X    | -    | X    | X    | X     | X     |

## Legend:

X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

## Canada

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

| Component | Canada - National Pollutant Release Inventory (NPRI) | Canadian Environmental Protection Agency (CEPA) - List of Toxic Substances | Canada's Chemicals Management Plan (CEPA) |
|-----------|------------------------------------------------------|----------------------------------------------------------------------------|-------------------------------------------|
| Zinc      | Part 1, Group A Substance                            |                                                                            |                                           |

## Other International Regulations

## Authorisation/Restrictions according to EU REACH

Not applicable

| Component | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-----------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Zinc      | -                                                                   | Use restricted. See item 75. (see link for restriction details)               | -                                                                                                     |

## Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component | CAS-No    | OECD HPV | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|-----------|-----------|----------|------------------------------|---------------------------|--------------------------------------------|
| Zinc      | 7440-66-6 | Listed   | Not applicable               | Not applicable            | Not applicable                             |

| Component | CAS-No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|-----------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Zinc      | 7440-66-6 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

## 16. Other information

Prepared By

Product Safety Department  
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www.thermofisher.com

**Creation Date** 09-May-2012  
**Revision Date** 26-March-2024  
**Print Date** 26-March-2024  
**Revision Summary** New emergency telephone response service provider.

**Disclaimer**

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**End of SDS**